Vani Seth

+1 (573) 968-9444 | vs4ky@missouri.edu
vaniseth | vaniseth | vaniseth | vaniseth

Education

- **Doctor of Philosophy in Computer Science** *University of Missouri* **Columbia, MO** *2025–Present*
- **Master of Science in Computer Science** *University of Missouri* **Columbia, MO** *2024–2025*
- **Bachelor of Technology in Computer Science** *Jaypee University* **Madhya Pradesh, India** *2020–2024*

Skills Summary

- **Languages**: Python, C, C++, SQL, Go, Cypher, JavaScript, HTML/CSS
- **Frameworks**: LangChain, HuggingFace, TensorFlow, PyTorch, Keras, Numpy, Pandas, Scikit-Learn, OpenCV, Regex
- **Databases**: MySQL, SQLite, MongoDB, Neo4j, Apache Spark, Apache Hadoop
- **Tools**: Kubernetes, Docker, GIT AWS, GCP, OpenAI, Streamlit, Linux
- **Visualization**: Photoshop, Illustrator, Matplotlib, MLflow

Experience

- **Graduate Research Assistant** *University of Missouri, Columbia* **Columbia, MO** *08/2024 – Present*
  - Designed domain-specific RAG and LLM systems for critical-domain applications such as finance, healthcare, education, science, enabling multi-hop reasoning, expert retrieval, and scientific query answering.
  - Built a trustworthy and secure LLM framework integrating prompt-injection defense, behavioral anomaly detection, and context-aware validation for critical-domain chatbots.
- **AI/ML Intern** *Airports Authority of India* **Delhi, India** *06/2023 – 07/2023*
  - Developed a LLM-powered chatbot for internal aviation related document interaction, improving employee access and workflow automation.
- **Research Assistant** *MISN Group, IIT Delhi* **New Delhi, India** *06/2022 – 07/2022*
  - Implemented Federated Learning models for healthcare AI to enable collaboration while preserving privacy.
  - Designed and evaluated incentive mechanisms to improve reliable participation across decentralized medical data providers.

Projects

- **Adaptive LLM Threat Response Framework (Adversarial LLM Prompt Attacks, Critical Application Security, Human AI Decision Support)**: Research framework for securing critical-domain applications against adversarial attacks using behavioral anomaly detection, context-aware prompt classifier, and threat memory. **Tech:** Python, PyTorch, DistilBERT, Isolation Forests, LangChain, Docker
- **Mizzou Cancer Registry RAG Chatbot (Trustworthy AI, Healthcare, NLP)**: Clinical decision-support system that answers cancer coding and registry queries using coding manuals with document-grounded responses, citation tracking, and hallucination detection for regulatory compliance. **Tech:** Python, LLMs, LangChain, HuggingFace, AWS
- **ARES-Vidura – Scientific RAG Agent (RAG, Scientific Reasoning, Fine-tuning, Human Feedback)**: LLM-based agent for carbon nanotube (CNT) research that performs multi-hop query decomposition, evidence-grounded retrieval, and human-in-the-loop validation to prevent hallucinations and improve scientific reliability. **Tech:** Python, Cypher, Neo4j, Vector Databases, Prompt Engineering, LLM-as-Judge
- **MLOps for AI-Based Scientific Workflows (AWS, MLflow, Reproducible AI Pipelines)**: Designed and delivered a cloud-native MLOps module for the Mizzou Cloud DevOps (MCD) platform, enabling reproducible AI workflows with automated experiment tracking, model versioning, and artifact management using MLflow on AWS. **Tech:** AWS EC2, MLflow, Python, PyTorch, Tensorflow, Cloud Monitoring

Publications

- M-TAP: Detecting Multi-Turn Jailbreaks via Activation Probes, ACM WWW 2027. *Under Review*
- ALTR: Adaptive Defense Framework Against Adversarial LLM Threats in Critical-Domain Applications, IEEE ICHMS 2026.
- Disentangling Complex Questions in LLMs via Multi-Hop Dependency Graphs, Proceedings of the 34th ACM CIKM 2025.
- EmpathAI: A Trustworthy and Secure Conversational Agent for Mental Healthcare, ACSAC HealthSec 2025

Achievements

- **Outstanding Graduate Poster Award (April 2026)** – Ranked top amongst graduate students participating from all engineering departments across University of Missouri
- **Agentic AI for Real-World Impact Hackathon (Feb 2022)** – Won the hackathon by building an AI-driven Campus Safety Infrastructure Analysis System that reimagines how universities approach safety.
- **Smart India Hackathon 2022 Finalist** – Ranked in the top 5 teams nationwide for an email spoofing detection solution, sponsored by the Ministry of External Affairs, India.